

## LECTURE 2 Supplement

### Topics

Mutual fund applications (NAV, Redemptions, Passthrough taxation),  
Trading applications, Buying on margin (Margin calls, interest and dividends), Short selling (Margin call and dividends)

- A mutual fund consists of 50,000 shares of Stock A, 100,000 shares of Stock B, and 200,000 shares of Stock C
- The fund issued 500,000 shares to fund these positions
- The fund borrowed \$200,000 against a credit line with a bank
- Price data:

| Stocks  | Price 09/02 @<br>10:00AM (EST) | Price 09/02 @<br>4:30PM (EST) |
|---------|--------------------------------|-------------------------------|
| Stock A | 105                            | 110                           |
| Stock B | 73                             | 78                            |
| Stock C | 54                             | 52                            |

### Questions:

- 1 If you plan to buy 100 shares of the fund on 09/02 at 10:00AM (EST), how much would you pay?
- 2 If an investor wants to redeem 10,000 of the mutual fund shares on 9/2 at 1:00PM (EST), what would the new NAV be?

- A mutual fund has just reduced its position in Apple stock from 300,000 shares to 100,000
- The price of a share of Apple when the mutual fund sold the stock was \$230
- The fund established the position five years ago when Apple was trading at \$150 per share
- The capital gains from the Apple position is taxable at the long-term capital gains rate of 15%
- The fund will passthrough the tax expense to its investors
- The fund has 500,000 shares outstanding

### Questions:

- 1 If you own 10,000 shares in this mutual fund, how much tax do you have to pay on the Apple position?

- The price of a share of Stock A is currently \$350 and you want to buy 100 shares on margin
- Your broker requires an initial margin of 55% and a maintenance margin of 26% on all stock margin loans
- The broker charges an interest of 5% per year on margin loans

### Questions:

- 1 How much can you borrow on margin from your broker?
- 2 What should be the percentage margin in your account in 4 months, if the share price of Stock A dropped to \$300 and the stock also paid a dividend of \$12 per share?
- 3 What is your effective annual return over the 4 month period?
- 4 How low would the share price have to drop in 4 months before you are issued a margin call from the broker?

- A share of Stock B is currently trading at \$250
- You plan to short sell 1000 shares of the stock
- Your broker requires an initial margin of 55% and a maintenance margin of 26% on all stock margin loans

### Questions:

- 1 What should be the percentage margin in your account in 3 months, if the share price of Stock B increased to \$300 and the stock also paid a dividend of \$10 per share?
- 2 What is your effective annual return over the 3 month period?
- 3 How high would the share price have to go in 3 months before you are issued a margin call from the broker?

- The Oct31 2026 US Treasury Note has a face value of \$1000 and an annual coupon rate of 5% paid semi-annually
- The bond trades today for 100.112% of face value (i.e., \$1001.12)
- Your broker requires an initial margin of 50% and a maintenance margin of 25%
- All margin loans carry an effective annual interest rate of 5.4%
- You want to buy 100 of these bonds on margin

### Questions:

- 1 If you were to borrow as much as you can, what is the initial margin needed to buy the 100 bonds?
- 2 What should be the percentage margin in your account in 4 months, if the bond price were to drop to 98% of face value and the bond made one coupon payment during this period?